2014 Team Answers

T-1. 4

T-2. $\left(\frac{1}{4}, \frac{1}{27}\right)$

T-3. $\frac{2}{7}\sqrt{53}$

T-4. 326

T-5. $\frac{43}{48}$

T-6. 3432

T-7. $\frac{45}{2}$

T-8. $2 + \pi$

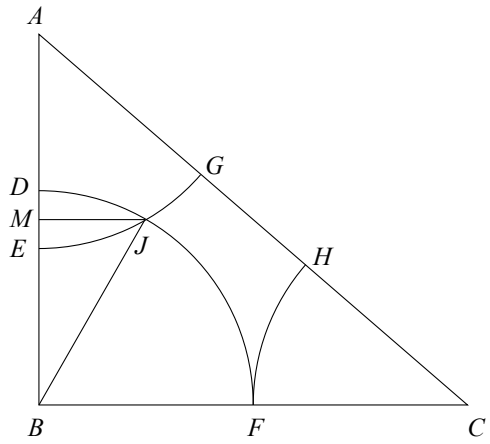
T-9. 10

T-10. $(-266, -3, -28)$

2014 Team Solutions

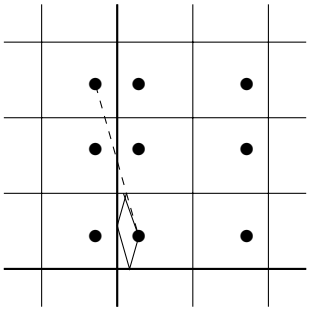
T-1. Consider the ordered pairs of digits (X, Y) for which $\underline{1}\,\underline{2}\,\underline{3}\,\underline{X}\,\underline{5}\,\underline{Y}\,\underline{7}$ **is** a multiple of 11. Recall that a number is a multiple of 11 if and only if the alternating sum of the digits is a multiple of 11. Because $1 + 3 + 5 + 7 = 16$, the sum of the remaining digits, namely $2 + X + Y$, must equal 5 or 16. Thus $X + Y$ must be either 3 or 14, making $X = 3 - Y$ (if $Y = 0$, 1, 2, or 3) or $14 - Y$ (if $Y = 5$, 6, 7, 8, or 9). Thus a solution (X, Y) exists unless $Y = \mathbf{4}$.

T-2. Label the triangle as $\triangle ABC$, with $AB = 2\sqrt{3}$ and $BC = 4$. Let D and E lie on $\overline{AB}$ such that $DB = AE = 2$. Let F be the midpoint of $\overline{BC}$, so that $BF = FC = 2$. Let G and H lie on $\overline{AC}$, with $AG = HC = 2$. Now draw the arcs of radius 2 between E and G, D and F, and F and H. Let the intersection of arc DF and arc EG be J. Finally, let M be the midpoint of $\overline{AB}$. The completed diagram is shown below.



The region R consisting of all points within $\triangle ABC$ that lie within 2 units of any vertex is the union of the three sectors EAG, DBF, and FCH. The angles of these sectors, being the angles $\angle A, \angle B$, and $\angle C$, sum to 180°, so the sum of their areas is 2π. Computing the area of R requires subtracting the areas of all intersections of the three sectors that make up R.
The only sectors that intersect are EAG and DBF. Half this area of intersection, the part above $\overline{MJ}$, equals the difference between the areas of sector DBJ and of $\triangle MBJ$. Triangle MBJ is a $1 : \sqrt{3} : 2$ right triangle because $BM = \sqrt{3}$ and $BJ = 2$, so the area of $\triangle MBJ$ is $\frac{\sqrt{3}}{2}$. Sector DBJ has area $\frac{1}{12}(4\pi) = \frac{\pi}{3}$, because $\text{m}\angle DBJ = 30°$. Therefore the area of intersection of the sectors is $2(\frac{\pi}{3} - \frac{\sqrt{3}}{2}) = \frac{2\pi}{3} - \sqrt{3}$. Hence the total area of R is $2\pi - (\frac{2\pi}{3} - \sqrt{3}) = \frac{4\pi}{3} + \sqrt{3}$. The total area of $\triangle ABC$ is $4\sqrt{3}$, therefore the desired probability is $\frac{\frac{4\pi}{3} + \sqrt{3}}{4\sqrt{3}} = \frac{\pi}{3\sqrt{3}} + \frac{1}{4}$. Then $a = \frac{1}{4}$ and $b = \left(\frac{1}{3\sqrt{3}}\right)^2 = \frac{1}{27}$, hence the answer is $\left(\mathbf{\frac{1}{4}}, \mathbf{\frac{1}{27}}\right)$.

T-3. Consider repeatedly reflecting square $ARML$ over its sides so that the entire plane is covered by copies of $ARML$. A path starting at $(2/7, 3/7)$ that touches one or more sides and returns to $(2/7, 3/7)$ corresponds to a straight line starting at $(2/7, 3/7)$ and ending at the image of $(2/7, 3/7)$ in one of the copies of $ARML$. To touch three sides, the path must cross three lines, at least one of which must be vertical and at least one of which must be horizontal.



If the path crosses two horizontal lines and the line $x = 0$, it will have traveled a distance of 2 units vertically and $4/7$ units vertically for a total distance of $\sqrt{2^2 + (4/7)^2}$ units. Similarly, the total distance traveled when crossing two horizontal lines and $x = 1$ is $\sqrt{2^2 + (10/7)^2}$, the total distance traveled when crossing two vertical lines and $y = 0$ is $\sqrt{2^2 + (6/7)^2}$, and the total distance traveled when crossing two vertical lines and $y = 1$ is $\sqrt{2^2 + (8/7)^2}$. The least of these is

$$\sqrt{2^2 + (4/7)^2} = \mathbf{\frac{2}{7}\sqrt{53}}.$$

T-4. If 306 is an element of S_k, then there exists an integer $m \geq 0$ such that $306 = k + mk^2$. Thus $k \mid 306$ and $k^2 \mid 306 - k$. The second relation can be rewritten as $k \mid 306/k - 1$, which implies that $k \leq \sqrt{306}$ unless $k = 306$. The prime factorization of 306 is $2 \cdot 3^2 \cdot 17$, so the set of factors of 306 less than $\sqrt{306}$ is $\{1, 2, 3, 6, 9, 17\}$. Check each in turn:

$$\begin{aligned} 306 - 1 = 305, &\quad 1^2 \mid 305 \\ 306 - 2 = 304, &\quad 2^2 \mid 304 \\ 306 - 3 = 303, &\quad 3^2 \nmid 303 \\ 306 - 6 = 300, &\quad 6^2 \nmid 300 \\ 306 - 9 = 297, &\quad 9^2 \nmid 297 \\ 306 - 17 = 289, &\quad 17^2 \mid 289. \end{aligned}$$

Thus the set of possible k is $\{1, 2, 17, 306\}$, and the sum is $1 + 2 + 17 + 306 = \mathbf{326}$.

T-5. Let $\log_x y = a$. Then the first equation is equivalent to $2a + \frac{5}{a} = 2k - 1$, and the second equation is equivalent to $\frac{5a}{2} - \frac{3}{2a} = k - 3$. Solving this system by eliminating k yields the quadratic equation $3a^2 + 5a - 8 = 0$, hence $a = 1$ or $a = -\frac{8}{3}$. Substituting each of these values

of a into either of the original equations and solving for k yields $(a,k) = (1,4)$ or $\left(-\frac{8}{3}, -\frac{149}{48}\right)$. Adding the values of k yields the answer of **43/48**.

Alternate Solution: In terms of $a = \log_x y$, the two equations become $2a + \frac{5}{a} = 2k - 1$ and $\frac{5a}{2} - \frac{3}{2a} = k - 3$. Eliminate $\frac{1}{a}$ to obtain $31a = 16k - 33$; substitute this into either of the original equations and clear denominators to get $96k^2 - 86k - 1192 = 0$. The sum of the two roots is $86/96 = \mathbf{43/48}$.

T-6. Define a *fault line* to be a side of a tile other than its base. Any tiling of $WASH$ can be represented as a sequence of tiles $t_1, t_2, \ldots, t_{14}$, where t_1 has a fault line of $\overline{WH}$, t_{14} has a fault line of $\overline{AS}$, and where t_k and t_{k+1} share a fault line for $1 \le k \le 13$. Also note that to determine the position of tile t_{k+1}, it is necessary and sufficient to know the fault line that t_{k+1} shares with t_k, as well as whether the base of t_{k+1} lies on $\overline{WA}$ (abbreviated "B" for "bottom") or on $\overline{SH}$ (abbreviated "T" for "top"). Because rectangle $WASH$ has width 7, precisely 7 of the 14 tiles must have their bases on $\overline{WA}$. Thus any permutation of 7 B's and 7 T's determines a unique tiling $t_1, t_2, \ldots, t_{14}$, and conversely, any tiling $t_1, t_2, \ldots, t_{14}$ corresponds to a unique permutation of 7 B's and 7 T's. Thus the answer is $\binom{14}{7} = \mathbf{3432}$.

Alternate Solution: Let $T(a,b)$ denote the number of ways to triangulate the polygon with vertices at $(0,0)$, $(b,0)$, $(a,1)$, $(0,1)$, where each triangle has area $1/2$ and vertices at lattice points. The problem is to compute $T(7,7)$. It is easy to see that $T(a,0) = T(0,b) = 1$ for all a and b. If a and b are both positive, then either one of the triangles includes the edge from $(a-1,1)$ to $(b,0)$ or one of the triangles includes the edge from $(a,1)$ to $(b-1,0)$, but not both. (In fact, as soon as there is an edge from $(a,1)$ to $(x,0)$ with $x < b$, there must be edges from $(a,1)$ to $(x',0)$ for all $x \le x' < b$.) If there is an edge from $(a-1,1)$ to $(b,0)$, then the number of ways to complete the triangulation is $T(a-1,b)$; if there is an edge from $(a,1)$ to $(b-1,0)$, then the number of ways to complete the triangulation is $T(a,b-1)$; thus $T(a,b) = T(a-1,b) + T(a,b-1)$. The recursion and the initial conditions describe Pascal's triangle, so $T(a,b) = \binom{a+b}{a}$. In particular, $T(7,7) = \binom{14}{7} = \mathbf{3432}$.

T-7. Because $\cos 2x = 1 - 2\sin^2 x$, $\sin^2 x = \dfrac{1-\cos 2x}{2}$. Thus the desired sum can be rewritten as

$$\frac{1-\cos 8^\circ}{2} + \frac{1-\cos 16^\circ}{2} + \cdots + \frac{1-\cos 352^\circ}{2} = \frac{44}{2} - \frac{1}{2}\left(\cos 8^\circ + \cos 16^\circ + \cdots + \cos 352^\circ\right).$$

If $\alpha = \cos 8^\circ + i \sin 8^\circ$, then α is a primitive 45th root of unity, and $1+\alpha+\alpha^2+\alpha^3+\cdots+\alpha^{44} = 0$. Hence $\alpha + \alpha^2 + \cdots + \alpha^{44} = -1$, and because the real part of α^n is simply $\cos 8n^\circ$,

$$\cos 8^\circ + \cos 16^\circ + \cdots + \cos 352^\circ = -1.$$

Thus the desired sum is $22 - (1/2)(-1) = \mathbf{45/2}$.

Alternate Solution: The problem asks to simplify the sum

$$\sin^2 a + \sin^2 2a + \sin^2 3a + \cdots + \sin^2 na$$

where $a = 4^\circ$ and $n = 44$. Because $\cos 2x = 1 - 2\sin^2 x$, $\sin^2 x = \dfrac{1-\cos 2x}{2}$. Thus the desired sum can be rewritten as

$$\frac{1-\cos 2a}{2} + \frac{1-\cos 4a}{2} + \cdots + \frac{1-\cos 2na}{2} = \frac{n}{2} - \frac{1}{2}\left(\cos 2a + \cos 4a + \cdots + \cos 2na\right).$$

Let $Q = \cos 2a + \cos 4a + \cdots + \cos 2na$. By the sum-to-product identity,

$$\begin{aligned} \sin 3a - \sin a &= 2\cos 2a \sin a, \\ \sin 5a - \sin 3a &= 2\cos 4a \sin a, \\ &\vdots \\ \sin(2n+1)a - \sin(2n-1)a &= 2\cos 2na \sin a. \end{aligned}$$

Thus

$$\begin{aligned} Q \cdot 2\sin a &= (\sin 3a - \sin a) + (\sin 5a - \sin 3a) + \cdots + (\sin(2n+1)a - \sin(2n-1)a) \\ &= \sin(2n+1)a - \sin a. \end{aligned}$$

With $a = 4^\circ$ and $n = 44$, the difference on the right side becomes $\sin 356^\circ - \sin 4^\circ$; note that the terms in this difference are opposites, because of the symmetry of the unit circle. Hence

$$\begin{aligned} Q \cdot 2\sin 4^\circ &= -2\sin 4^\circ, \text{ and} \\ Q &= -1. \end{aligned}$$

Thus the original sum becomes $44/2 - (1/2)(-1) = \mathbf{45/2}$.

T-8. Call the region R, and let R_q be the portion of R in the qth quadrant. Noting that the point (x, y) is in R if and only if $(\pm x, \pm y)$ is in R, it follows that $[R_1] = [R_2] = [R_3] = [R_4]$, and so $[R] = 4[R_1]$. So it suffices to determine $[R_1]$.

In the first quadrant, the boundary equation is just $x^2 + y^2 = x + y \Rightarrow (x - \frac{1}{2})^2 + (y - \frac{1}{2})^2 = \frac{1}{2}$. This equation describes a circle of radius $\frac{\sqrt{2}}{2}$ centered at $(\frac{1}{2}, \frac{1}{2})$. The portion of the circle's interior which is inside the first quadrant can be decomposed into a right isosceles triangle with side length 1 and half a circle of radius $\frac{\sqrt{2}}{2}$. Thus $[R_1] = \frac{1}{2} + \frac{\pi}{4}$, hence $[R] = \mathbf{2 + \pi}$.

T-9. Write $a_n = a_1 + r(n-1)$ and $b_n = b_1 + s(n-1)$. Then $a_{20} + b_{14} = a_1 + b_1 + 19r + 13s$, while $b_{20} + a_{14} = a_1 + b_1 + 13r + 19s = a_{20} + b_{14} + 6(s - r)$. Because both sequences consist only of integers, r and s must be integers, so $b_{20} + a_{14} \equiv a_{20} + b_{14} \bmod 6$. Thus the least possible value of $b_{20} + a_{14}$ is 4. If $b_{20} = 3$ and $a_{14} = 1$, then $\{a_n\}$ must be a decreasing sequence (else a_{13} would not be positive) and $a_{20} \leq -5$, which is impossible. The case $b_{20} = a_{14} = 2$ violates the requirement that the terms be distinct, and by reasoning analogous to the first case, $b_{20} = 1, a_{14} = 3$ is also impossible. Hence the sum $b_{20} + a_{14}$ is at least 10. To show that 10 is attainable, make $\{a_n\}$ decreasing and b_{20} as small as possible: set $b_{20} = 1$, $a_{14} = 9$, and $a_n = 23 - n$. Then $a_{20} = 3$, yielding $b_{14} = 997$. Hence $s = \dfrac{997 - 1}{14 - 20} = \dfrac{996}{-6} = -166$ and

$b_1 = 997 - (13)(-166) = 3155$, yielding $b_n = 3155 - 166(n-1)$. Because $b_{20} = 1 \leq a_{20}$ and $b_{19} = 167 \geq a_1$, the sequences $\{b_n\}$ and $\{a_n\}$ are distinct for $1 \leq n \leq 20$, completing the proof. Hence the minimum possible value of $b_{20} + a_{14}$ is **10**. [Note: This solution, which improves on the authors' original solution, is due to Ravi Jagadeesan of Phillips Exeter Academy.]

T-10. First, eliminate x: $y(y^2z - x) + (xy - z^2) = 14(y+1) \Rightarrow z^2 - y^3z + 14(y+1) = 0$. Viewed as a quadratic in z, this equation implies $z = \dfrac{y^3 \pm \sqrt{y^6 - 56(y+1)}}{2}$. In order for z to be an integer, the discriminant must be a perfect square. Because $y^6 = (y^3)^2$ and $(y^3-1)^2 = y^6 - 2y^3 + 1$, it follows that $|56(y+1)| \geq 2|y^3| - 1$. This inequality only holds for $|y| \leq 5$. Within that range, the only values of y for which $y^6 - 56y - 56$ is a perfect square are -1 and -3. If $y = -1$, then $z = -1$ or $z = 0$. If $y = -3$, then $z = 1$ or $z = -28$. After solving for the respective values of x in the various cases, the four lattice points satisfying the system are $(-15, -1, -1), (-14, -1, 0), (-5, -3, 1)$, and $(-266, -3, -28)$. The farthest solution point from the origin is therefore **$(-266, -3, -28)$**.